Wings Intelligent Load Balancer — User Manual

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1. Introduction

Wings Load Balancer lets every user open one address, sign in with their company and username, and land on the application server that holds their company's data. Nobody needs to know a server name.

This manual covers the two things people use:

Part	Who uses it	Address (local trial)
Sign-in gateway — the public entry point; it signs users in and forwards every request to the right server	Everyone who uses Wings applications	http://localhost:5280
Admin console — where servers, companies, user assignments, pools and policies are managed and monitored	Administrators, operators, auditors	http://localhost:5290

In production the gateway sits behind your public address (for example `https://erp.example.com`) and the admin console on an internal address.

Five terms appear throughout:

- **Server** — an application server running a Wings application.
- **Company** — a tenant. Users always belong to a company, and the same username can exist in two companies as two different people.
- **Assignment** — a rule that says which server a user or a company uses. Assignments can start and end on set dates.
- **Pool** — a group of servers that companies can share; the gateway spreads users across the healthy ones.
- **Failover policy** — the list of servers a tenant may move to when its own server is unavailable. Without one, the tenant waits rather than being sent somewhere unauthorised.

Sections 2 and 3 are for everyone; sections 4–11 walk through each admin page; section 12 explains how the gateway decides; sections 13–14 give step-by-step tasks and fixes.

2. For end users: signing in

You sign in once at your organisation's Wings address, and every page after that comes from your organisation's server automatically.

1. Open your Wings address. If you are not signed in, you see the **Sign in** page; a link to a specific page brings you back to that page after sign-in.
2. Enter **Company**: your company code or company name (for example `ABC` or `ABC Motors`). Upper or lower case and extra spaces do not matter.
3. Enter your **Username** and **Password**, then select **Sign in**.
4. If your account uses two-factor authentication, an **Authentication code** box appears. Enter the 6-digit code from your authenticator app and select **Sign in** again.
5. If your company signs in through a corporate identity provider, you are sent to that provider's page instead of seeing a password box.

Signing out: use your application's sign-out link, or open `/_wings/logout` on your Wings address. Your session also ends after 30 minutes without activity, and after 12 hours in any case.

Messages you may see

Message	What it means	What to do
The company, username or password is incorrect.	Any sign-in detail is wrong, or the account is disabled or locked. The message never says which, to protect accounts.	Check the details. After 5 failed attempts the account is locked for 15 minutes; ask your administrator if it stays locked.
Your organisation has reached its concurrent user limit.	Your company's maximum number of users is already signed in.	Try again later, or ask your administrator to raise the limit.
Sign-in is temporarily unavailable.	The sign-in service cannot reach its user directory.	Try again in a few minutes. People already signed in are not affected.
Service temporarily unavailable for your organisation	Your company's server is down and no approved backup server is available.	Try again shortly. Quote the **Reference** shown on the page when you contact support.
Access denied	Your account no longer has access to this application. You are signed out.	Contact your administrator.

Message	What it means	What to do
Application unavailable	The server stopped responding in the middle of your request.	Try again; if it repeats, contact support with the **Reference**.

3. Admin console basics

The admin console is a separate site with its own administrator accounts; end-user accounts cannot sign in to it.

Signing in

1. Open the admin console address and enter your administrator **Username** and **Password**.
2. If two-factor authentication is on for your account, enter the 6-digit code when asked.
3. You land on the first page your role allows, usually the **Dashboard**. After 60 minutes without activity you are signed out.

Five failed attempts lock the account for 15 minutes. A disabled administrator is signed out within 5 minutes, even mid-session.

Your account

Open the user menu (your name, top right):

- **Change password** — enter the current password and a new one of at least 12 characters.
- **Two-factor authentication** — select **Generate secret**, add the setup link or manual key to your authenticator app, then enter the 6-digit code it shows and select **Enable**. If two-factor is already on, you must first enter a code from your current authenticator; two-factor sign-in is then off until you confirm a code from the new secret, so finish the setup in one go.
- **Sign out**.

Finding your way around

The top bar shows a green **Live** dot while real-time updates are connected, and a bell with the number of open alerts. The side menu (toggle it with the menu button, top left) groups the pages:

Group	Pages
Monitor	Dashboard, Alerts, Active sessions
Infrastructure	Servers, Server pools, Maintenance
Tenants	Companies, User assignments, Bulk upload
Routing	Policies & applications, Routing simulator, Routing history, Configuration
Governance	Audit log, Reports, Settings, Administrators

You see only the pages and buttons your role allows; anything else is hidden or greyed out.

Roles

Role	Can do
SuperAdministrator	Everything, including settings, administrators and exporting backend keys
RoutingAdministrator	Manage servers, companies, users, assignments, pools, policies; operate servers; simulate; read audit, alerts, reports; revoke sessions
Operator	View everything operational; enable, disable, drain and restore servers; simulate routing; handle alerts; revoke sessions
ReadOnlyMonitor	View servers, companies, assignments, pools, dashboards, alerts, reports and sessions; change nothing
Auditor	Read the audit log, dashboards, reports, servers, companies and assignments

Habits that help

- **Reasons.** Most changes ask for an optional reason. It is stored in the audit log next to your name, so a short "why" saves questions later.
- **"Changed by someone else" messages.** If two administrators edit the same record, the second save is refused instead of silently overwriting. Reload the page and make your change again.

- **Changes go live by themselves.** Saved routing changes reach every gateway within about 15 seconds; nothing needs restarting.

4. Dashboard

The Dashboard answers "is everything routing normally right now?" at a glance, and updates live while the **Live** dot is green. Select **Refresh** to reload everything at once.

Summary tiles

Tile	Shows	Worth a look when
Servers	Registered servers, with draining and unknown counts	Unknown stays above 0 for minutes (no health data)
Healthy / Degraded / Offline	Servers in each state. Offline includes disabled, in maintenance or failing health checks	Degraded or Offline is above 0
Companies	Companies registered	—
Dedicated vs shared	Companies with a dedicated route / a shared-pool route	—
Users on dedicated servers	Users with their own server assignment	—
Active sessions	Signed-in sessions active in the last 30 minutes	It drops suddenly
Requests / sec	Traffic across all live gateways	It drops to 0 during working hours
Failed routing (1 h)	Requests refused in the last hour (no healthy server, no assignment, access denied)	Above 0
Avg gateway latency	Average time per proxied request, plus the routing decision time	It climbs steadily
Current failovers	Sessions currently running on a backup server	Above 0
Open alerts	Unresolved alerts	Above 0

Traffic and routing decisions

Charts of routing decisions (routed vs refused) and requests processed over time. Narrow them with **Company**, **Server**, **Pool**, **Application**, **Environment**, **From** and **To**, and choose the **Bucket** size; **Clear** resets the filters. Each chart has **View as table** for the exact numbers.

Server utilisation

One row per server with health, type, CPU, memory, active requests and response time, each with a small trend line. Choose the window with **1 h**, **6 h**, **24 h** or **72 h**. Select a server name to open its page.

Live routing decisions

A running feed of new sessions, reroutes, failovers and refusals as gateways make them, for example *ABC → APP-SERVER-01 · New session · User assignment*. **Full history** opens Routing history.

Topology

A tree of **Environment → pool or dedicated group → server → company**, with each server's health. Use it to answer "which companies are on this server?". Select a server or company to open it; the two buttons expand or collapse the whole tree.

Gateway instances

Each gateway with its status, requests per second, latency, active requests, whether its configuration is **In sync**, and the state of its database and cache connections. A gateway that is not in sync, or shows a dependency warning, is still routing on its last good configuration; see section 14.

5. Servers

Servers lists every application server with its status and health; select a name to open its page, where you edit it and run every server action.

Status at a glance

Each server shows two badges: its **status** (what you decided) and its **health** (what the health checks found).

Status	Meaning	Takes new users?	Keeps existing users?
Online	Active and passing health checks	Yes	Yes
Degraded	Active but slow, overloaded or reporting errors	Yes, by default	Yes
Maintenance (draining)	Being emptied before maintenance	No	Yes, until they finish or the drain timeout
Maintenance	In maintenance	No	No; they move to an approved backup
Offline	Disabled, or failing health checks	No	No; they move to an approved backup
Retired	Out of service for good	No	No

Health moves Healthy → Degraded → Unhealthy → Recovering → Healthy. A recovering server returns to service only after passing checks for a steady period (30 seconds by default).

Registering a server

1. Select **Add server**.
2. Fill in the details (fields below) and select **Save**.
3. Open the new server and select **Test connectivity**. A pass shows the response time and the metrics the server reported.
4. Add the server to a pool, or assign companies or users to it (sections 6–8). A new server takes no traffic until something points to it.

Field	What to enter
Name	Unique short name, for example `APP-SERVER-08`
Hostname, IP address	The server's internal DNS name; IP is optional
Base URL (backend)	Address the gateway forwards to. Must match the allowed backend hosts or networks set by IT, and use https in production

Field	What to enter
Health check URL	Usually `<base URL>/health/ready`; must be on the same host
Public URL (redirect mode)	Only for applications that redirect users straight to the server
Environment, Region	Production, Staging, Test or Development; the site or data centre
Server type	**Dedicated** (one tenant only), **Shared** (pools), or **Hybrid** (both)
CPU cores, RAM, Max capacity	Size, and the most sessions it should hold (0 = unlimited)
Application version, Hosted applications	Which Wings applications it runs, and at what version
Priority, Weight	Weight is its share of pool traffic: 200 gets twice as much as 100
Drain timeout	Minutes to wait for users to finish when draining (default 30)
Require manual restore after recovering from failure	Keep it out of service after an outage until someone selects Restore
Transport security	Optional client-certificate and pinned-certificate thumbprints, supplied by IT

Server actions (the Operate menu)

Action	Use it to
Drain…	Stop new users, let current users finish, then move the server to Maintenance automatically. Enter a timeout and reason
Maintenance	Take it out of service now. Current users move to an approved backup
Restore to service	Return it to Active. Also clears a manual-restore hold
Disable	Switch it off for routing (for example a fault you are investigating)

Action	Use it to
Enable	Switch a disabled server back on

The ⋮ menu holds the rarer actions:

- **Retire…** removes the server permanently. It is refused while any company, user or pool still uses it, and the message lists what to move first.
- **Delete…** is only for a server that was never used; otherwise retire it.
- **Backend configuration…** shows the server's **Signing key** and **Health key**, which the application team installs on the server. It needs special permission, is logged, and the keys should be copied straight into the server's secret store.

Server page tabs

- **Overview** — live status (health, last check, CPU, memory, response time, sessions against capacity, pools) and the edit form.
- **Assignments** — companies and users using it, its pools, and **capacity reservations**: sessions held back for an important company. Use **Set reservation**; 0 removes one.
- **Active sessions** — who is on it now.
- **Performance** — CPU, memory, requests, response time and errors over time.
- **Health history** — every health change with its reason.
- **Audit history** — every change made to it, by whom and why.

6. Companies

A company's page shows where its users go today and is where you change that: a dedicated server, a cluster of servers, or a shared pool.

Adding a company

1. On **Companies**, select **Add company**.
2. Enter the **Company code** (what users type at sign-in, for example `ABC`) and **Name**.
3. Add **Aliases**: other names users may type, such as `ABC Motors` or `ABC-MOTORS`. An alias cannot be used by two companies.
4. Set **Service plan**, **Default environment**, **Max concurrent users** (blank = no limit) and **Authentication** (local passwords, the Wings authentication service, or a corporate identity provider).
5. Select **Save**. The company cannot sign in to anything until it has a route (next step).

Routing mode on the form is a label for reports; the actual route comes from the assignments below.

Giving a company a route

Open the company, go to **Assignments**, and choose one:

- **Assign / change server…** for a dedicated server. Pick the **Primary server**, optionally a **Secondary server** and **Failover policy**, the **Environment**, and optionally one **Application** (blank = all applications).
  - For a cluster, also pick a **Cluster pool** and **Cluster mode**: *PrimaryPreferred* uses the primary and keeps the others as backups; *Balanced* spreads users across all cluster members.
  - Select **Preview** to see which server sample users would get before and after, then **Apply now**, or set **Effective from** to **Schedule change**.
- **Assign shared pool…** for shared servers. Pick the **Pool**, **Environment** and **Priority** (lower is tried first if the company has several pools). Tick *End the company's dedicated assignment in this scope* to switch from dedicated to shared.

Assignments are never overwritten: the old one gets an end date, so the table shows what is **Current**, **Scheduled** and **Ended**. Use **End** to stop one.

Two safety rules apply. A **Dedicated** server serves one company only, so you cannot give it to a second company. And a server must be in the company's environment and run the chosen application.

Data authorisations

This tab lists the servers allowed to hold the company's data. Servers named in the company's own assignments, cluster, failover pool and shared pools are allowed automatically. Use **Grant server** to allow one more, for example before giving one user a server of their own (section 7); **Revoke** removes a grant that no user still depends on.

Transferring a company to another server

1. Select **Transfer…** and choose the **Target server** (and optionally a **Target secondary server**).
2. Select **1. Validate**. It checks environment and application match, tenant isolation, the target's health, whether it accepts new users, application version and capacity. Every check must pass.
3. Choose the cutover:
   - **Immediate** — every session moves to the target on its next request.
   - **Scheduled** — the switch happens at **Cutover at** (pick a quiet time).

- **Force re-authentication** — switch now and sign everyone out.

4. Select **2. Perform transfer**, then confirm with **Simulate routing**.

Disabling access

Disable access… suspends or disables the company; its users are refused at their next request. Tick the option to sign everyone out at once. **Enable access…** reverses it.

Other tabs

Overview (edit details, routing summary), **Users** (members; **Manage user assignments** jumps to section 7), **Sessions** (who is signed in now) and **History** (every change to the company).

7. User assignments

A user assignment gives one user their own server, overriding their company's route; users without one simply follow the company.

Finding a user

User assignments lists one row per company membership: **Company**, **Username**, **Assigned server** and **Routing type**, which is *User Dedicated*, *Company Default*, *Shared Pool* or *Unassigned*. Filter by **Company**, search by **Username or display name**, or tick **Only users with a dedicated server**.

The same username in two companies is two different people with separate rows; changing one never affects the other.

Select a row to open the user's panel underneath.

Assigning a dedicated server

1. In the panel, under **Assign dedicated server**, choose the **Environment** and, if it applies to one application only, the **Application**.
2. Choose the **Dedicated (primary) server**. Only servers authorised for the company's data can be picked; the others are greyed out with "not authorised". To use one of those, grant it on the company's **Data authorisations** tab first (section 6).
3. Optionally choose a **Secondary server** and a **Failover policy**, or tick **Fall back to the company route if unavailable**. With none of these, the user waits if their server is down rather than being moved.
4. Leave **Effective from** empty to apply now, or set a future date and time to schedule the change. **Effective to** ends it automatically.

5. Enter a **Reason**, select **Preview** to see the user's server before and after (nothing is saved), then **Assign now** (or **Schedule assignment**).

The gateway picks up the change within about 15 seconds. From then on the user's requests, including those of a session already open, go to the new server.

Changing or undoing

- **Current and scheduled user-level assignments** lists what applies; **Edit** changes one, **Remove override** returns the user to the company route.
- **History** lists every past assignment with who changed it and why. **Rollback** restores an earlier one as a new current assignment.

Managing the person

- **Create user** adds a user to a company: **Company**, **Username**, **Display name**, **E-mail**, and either an **Initial password** (at least 12 characters) or an **External subject** for users who sign in through the Wings authentication service or an identity provider.
- **Disable membership** blocks the user from this company at their next request; **Enable membership** reverses it.
- **Reset password** sets a new password and clears any lockout.
- **Simulate** opens the Routing simulator for this user (section 9).

Bulk upload

Use **Bulk upload** to assign or remove many users at once from a CSV or Excel (.xlsx) file.

1. **Download template** and fill one row per change:

Column	Required	Example
Company	Yes	`ABC`
Username	Yes	`user1`
Action	No (default Assign)	`Assign` or `Remove`
PrimaryServer	For Assign	`APP-SERVER-01`
SecondaryServer	No	`APP-SERVER-02`
FailoverPolicy	No	`No failover`
Application	No	`HR`
Environment	No (default Production)	`Production`

Column	Required	Example
EffectiveFrom, EffectiveTo	No	`2026-12-01T00:00:00Z`
Reason	No	`Finance team move`

2. Upload the file and select **Validate**. The **Dry-run results** show every row with its errors, for example an unknown user or a server not authorised for that company.
3. Fix the file and validate again until every row is valid. **Commit** is only enabled then, and it applies all rows together or none. One upload holds up to 5,000 rows.

8. Pools, policies and applications

Pools group servers that tenants share; policies decide how users are spread and where they may fail over; applications tell the gateway which public address belongs to which Wings application.

Server pools

On **Server pools**, select **Create pool** and set:

- **Name**, **Environment**, **Region**, **Description**.
- **Load-balancing algorithm** and **Session affinity** (tables below).
- **Failover policy** — where users go if every member is down. Without one, they wait.
- **Eligible service plans** — limit the pool to companies on certain plans; empty means any company assigned to it.
- **Applications** the pool serves, and **Max capacity (sessions)**.

Use **Add member** to add servers. Each member has a **Weight override** (blank = the server's own weight), a **Priority** (lower is tried first for failover) and a switch that stops it taking new users while keeping current ones — a quick way to ease one server out without draining it. A server can be in several pools; a **Dedicated** server cannot join a pool that companies share.

Algorithm	Picks
RoundRobin	Each server in turn
WeightedRoundRobin	In turn, in proportion to weight
LeastRequests	The server with the fewest requests in progress
WeightedLeastRequests	Fewest requests relative to weight

Algorithm	Picks
LowestUtilization	The lowest load score (CPU, memory, requests, response time, errors)
ConsistentHashCompany	Always the same server for a company
ConsistentHashCompanyUser	Always the same server for a user

Session affinity	Effect
None	Every request is balanced afresh; use only for applications that keep no state on the server
PerSession	A session stays on the server it started on
PerUser	A user always lands on the same server, from any gateway
PerCompany	All users of a company land on the same server

With any affinity except None, a user stays on their server while it remains healthy and permitted; they move only if it fails, drains to maintenance or leaves the pool.

Failover policies

Under **Policies & applications → Failover policies**, select **New policy**:

- **Failover enabled** — off means "never move these users"; they get a temporary-unavailable page instead.
- **Failover pool** — the approved backup servers. The assignment's secondary server, if any, is tried first.
- **Max failover attempts** — how many backups to try (0 = all).
- **Failback** — what happens when the original server recovers: **Automatic** moves users back straight away; **EndOfSession** leaves them on the backup until they sign out; **Manual** does the same, and you pair it with *Require manual restore* on the server so it waits for you.

A backup is never used if it is dedicated to another company, even when it is listed in the pool.

Routing policies

Routing policies tune how the gateway judges servers. The policy marked **Default policy** applies to every company without its own.

- **Allow routing to degraded servers** — on by default; turn off to treat Degraded like Offline.
- **Load-score metric weights** — how much **CPU**, **Memory**, **Active requests**, **Response time** and **Error rate** count when the LowestUtilization algorithm or failover ordering compares servers. The default split is 30 / 20 / 25 / 15 / 10%; **Normalise to 1.00** rescales your numbers. **Response-time ceiling (ms)** is the response time counted as fully loaded.

Default affinity and the session timeouts on this form are recorded for reference only: pools set their own affinity, and the gateway's configuration sets session timeouts (30 minutes idle, 12 hours maximum).

Applications and environment bindings

Each Wings application (for example ERP and HR) has a **Code**, **Name**, **Minimum server version** (servers running an older version are skipped) and **Enabled** switch. Under it, **Add binding** connects a public address to the application in one environment:

Field	Example	Meaning
Public host	`erp.example.com`	The address users open
Path prefix	`/hr`	Optional; `/` for the whole host. The longest matching prefix wins
Environment	Production	Users arriving here are routed only to servers in this environment
Routing mode	Proxy	**Proxy** keeps users on the public address (recommended); **Redirect** sends them to the server's public URL
Request timeout (s), Max request body (MB)	100, 200	Limits for this application

9. Routing simulator, history and configuration

The **Routing simulator** shows which server a user would get and why, using the live configuration and health, without changing anything or signing anyone in.

Running a simulation

1. Enter **Company (code, name or alias)** and **Username**, pick the **Application** and **Environment**, and select **Simulate**.
2. Read the result card:

Panel	Tells you
Headline	*Routed to APP-SERVER-02*, or why not (for example *No healthy destination*), with the reason and configuration version
Resolved identity	The company and user the names resolved to, membership status and application
Decision	Which rule applied (user assignment, company assignment, shared pool), the chosen server and its health, the **failover server** that would take over, the routing and failover policy
User assignment / Company assignment	The rule found at each level, with its dates
Eligible shared pools	Pools the company may use
Step-by-step trace	Each check in order, including servers skipped and why

Use it before and after any change, and first thing when a user reports landing on the wrong server.

End-to-end test

The **End-to-end test** goes further: it really signs in through the gateway with a test account and makes one request. Enter **Company**, **Username**, **Password**, optionally a **Path** (for example `/api/whoami`) and a **Host header** for another environment, then **Run test**. The result shows each step's status and time, and which server answered when the application reports it. The password is used once and never stored or shown again; use a test account, not a real person's.

Routing history

Routing history lists the decisions gateways recorded: new sessions, reroutes, failovers, failbacks and refusals. Filter by company, username, server, pool, application, environment, date range, **Event** and **Outcome**. Every request is not logged — only these changes — so a quiet history is normal.

Configuration

Configuration shows the **Current version** of the routing configuration, the **Gateway sync status** (each gateway's version, and whether it is in sync), any **Validation issues** and the **Version log** of recent changes.

Treat validation **errors** as urgent: they include a dedicated server shared by two companies, an assignment pointing to a server in the wrong environment, and references to a missing server, pool or policy. Warnings, such as an active company with no route, a pool with no members or an assignment to a disabled server, deserve a look but may be intentional.

10. Maintenance, sessions and alerts

These three pages cover day-to-day operation: planned downtime, who is signed in, and what needs attention.

Maintenance windows

Schedule downtime in advance instead of draining by hand on the day.

1. On **Maintenance**, select **Schedule maintenance**.
2. Choose the **Server**, when it **Starts** and **Ends**, the **Drain timeout (minutes)** and a **Reason**.
3. Leave **Return the server to service automatically when the window ends** ticked to have it come back on its own. It returns only once its health checks pass; untick it to restore it yourself.
4. Select **Schedule**.

At the start time the server begins draining, moves to maintenance when users are gone (or the timeout passes) and, at the end, returns to service. Each step appears in the audit log. A window that has not started can be **Cancel**led; once it is under way, use **Restore to service** on the server instead. Two windows for one server cannot overlap. **Include completed and cancelled** shows past windows.

Active sessions

Active sessions lists everyone active in the last 30 minutes: company, username, application, server, gateway, start and last-seen times, and a masked network address. Filter by **Company**, **Server** or **Username contains**.

Revoke signs that session out on every gateway within about 2 seconds; the user must sign in again. To lock someone out for good, also disable their membership (section 7).

Alerts

Alerts lists problems the system has detected, newest and most severe first. Filter by **Status** (default: open and acknowledged) and minimum severity.

Alert	Raised when
Server unavailable (Critical)	A server fails its health checks
No healthy destination (Critical)	A company's requests were refused in the last 5 minutes
High CPU / memory / error rate, slow response	A server crosses the thresholds in **Settings**
Near capacity	Sessions reach the capacity threshold (85% by default)
Unexpected failover	A company failed over while its server was not being drained
Gateway connectivity	A gateway stopped reporting, or cannot reach the database
Redis unavailable	The shared cache is down; gateways continue on their own memory
Configuration inconsistency	Validation errors, or a gateway running on stale configuration

The same problem is counted, not repeated: one alert shows how many times it occurred. **Acknowledge** tells colleagues you are on it; **Resolve** closes it with a note. Most alerts also close themselves when the condition clears.

Notification channels

Under **Notification channels**, select **New channel** to send alerts out:

Type	Target	Notes
Email	Addresses separated by commas	Needs the mail server set up by IT
Webhook	An https URL	Each message is signed so the receiver can check it came from Wings
Teams	A Microsoft Teams incoming-webhook URL	Posts a card to the channel

Set the **Minimum severity** (Info, Warning or Critical) and select **Send test** to check delivery.

11. Audit log, reports, settings and administrators

Every change anyone makes is recorded, and every operational question has a report you can export to CSV or Excel.

Audit log

The **Audit log** records who did what, when, to which company, user or server, the value before and after, and the reason given. Automatic actions are recorded too: health changes and drains under *system* or *drain-controller*, failovers under the gateway's name. Passwords and secrets are never shown.

Narrow it with **Event type**, **Actor contains**, **Company**, **Server**, **User ID**, **From**, **To** and **Search reason / new value**, then **Apply** (**Clear** resets). Expand a row to see the full before-and-after values. **CSV** and **Excel** export what the filters show, up to 100,000 rows.

Useful event types: *UserAssignmentChanged*, *CompanyAssignmentChanged*, *CompanyTransferred*, *ServerDrainStarted*, *ServerDrainCompleted*, *ServerRestored*, *HealthTransition*, *AutomaticFailover*, *UnauthorizedRoutingAttempt*, *UserLoginFailed*, *AdministratorLogin*, *SessionRevoked*, *SecretsExported*.

Entries are kept for 365 days by default (see Settings).

Reports

Pick a report on the left, set **From** and **To** (default: the last 7 days), and select **Run**. **Filter rows** searches the result; **CSV** and **Excel** download it. Times are in UTC.

Report	Answers
Server utilisation over time	Hourly CPU, memory, requests, response time and errors per server
Company-wise server allocation	Each company's routing mode, servers, pool and failover policy
User-wise server allocation	Every current and scheduled user assignment
Routing decision history	Decisions logged by the gateways
Failover history	Every failover and failback

Report	Answers
Server downtime	Incidents, minutes down and availability % per server
Peak concurrent sessions	The busiest moment per server per day
Requests processed	Requests, refusals and failovers per gateway per hour
Latency distribution	How many responses fell in each time band
Application error rates	Hourly error rate per server
Capacity trends	Daily average and peak sessions against capacity

Settings

Settings holds the numbers the system works to. Change a value and select **Save**; **Discard changes** undoes unsaved edits.

Setting	Default	Controls
Retention: audit / decision log / metrics / health history / sessions / alerts	365 / 30 / 90 / 180 / 30 / 180 days	How long records are kept before automatic clean-up
Alerts: CPU / memory / error rate	85% / 90% / 5%	When a high-usage alert is raised
Alerts: response time	2,000 ms	When a slow-response alert is raised
Alerts: capacity	85%	When a near-capacity alert is raised
Sessions: idle minutes	30	How long a session counts as active without activity

Administrators

Administrators (SuperAdministrator only) has three tabs:

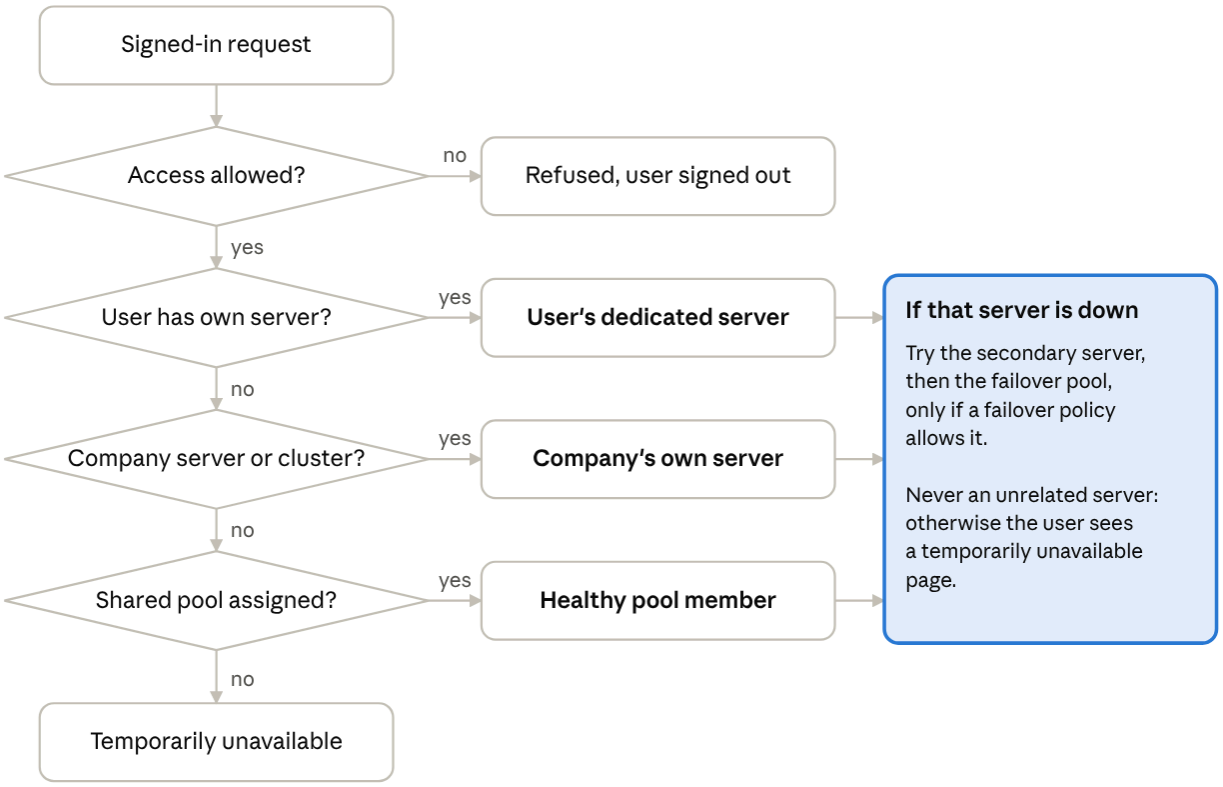
- **Administrators** — select **New administrator**, enter **Username**, **Display name**, **E-mail**, an **Initial password** (at least 12 characters), **Status** and **Roles**. **Edit** changes roles or status; leave the password empty to keep it. Set **Status** to Disabled to remove access; you cannot disable your own account.
- **Roles** — what each built-in role includes (section 3).

- **Permission catalogue** — every individual permission and what it allows.

12. How routing decides

For every request the gateway checks the user's own assignment first, then the company's, then the company's shared pools, and only ever sends the user to a server one of those rules permits.

Every request goes only to a server one of its rules permits



routing decision · 4 checks and the failover rule

Read it top to bottom: the first rule that matches decides, and a server that is down is replaced only by servers its failover policy names.

The order of checks

1. **Is access allowed?** The company must be active, the user's membership enabled, and the application published at the address used. If not, the request is refused and the user is signed out.
2. **User assignment.** If the user has their own server for this environment (and application), it is used.
3. **Company assignment.** Otherwise the company's dedicated server or cluster is used.

4. **Shared pool.** Otherwise the company's assigned pools are tried in priority order, and the pool's algorithm picks a healthy member.
5. **No rule at all** means the request is refused with a temporary-unavailable page. A company is never sent to "any available server".

At every step a candidate server is skipped if it is in the wrong environment, does not run the application (or runs too old a version), is disabled, retired or in maintenance, is unhealthy, is full, or is dedicated to another company. A draining server is skipped for new users only.

When the chosen server is unavailable

- With a **failover policy**, the user goes to the assignment's secondary server, then the policy's failover pool, in that order. When the original server recovers, the **Failback** setting decides whether they move back (section 8).
- **Without** one, the user sees a temporary-unavailable page until the server is back. This is deliberate: their data may exist only on that server.
- A user assignment can instead allow **Fall back to the company route**.

If a server stops answering in the middle of a simple page request, the gateway retries once on a permitted backup straight away; the user notices nothing. Uploads and form submissions are never retried automatically.

Staying on one server

Once a session starts on a server it stays there while that server stays healthy and permitted, so the application never loses track of the user. Pools with *PerUser* or *PerCompany* affinity go further: the same user or company lands on the same server from any gateway.

When parts of the system are down

What is down	What users notice	What to do
One application server	Users with an approved backup move to it; others see a temporary-unavailable page	Restore the server or attach a failover policy
One gateway	Nothing; the network load balancer sends them to another gateway and their session still works	Restart the gateway

What is down	What users notice	What to do
The routing database	Signed-in users keep working on the last known configuration for up to 30 minutes; new sign-ins wait	Restore the database. After 30 minutes the gateways stop routing rather than use stale rules
The shared cache (Redis)	Nothing, in most cases; a user may occasionally land on a different shared-pool server	Restore Redis; nothing needs restarting
The health monitor	Nothing at first; after 2 minutes without health updates, servers count as unknown and stop taking users	Start the standby monitor (it normally takes over by itself)

13. Common tasks, step by step

Each recipe ends with a check in the Routing simulator, which shows the result without affecting anyone.

Onboard a new company on a shared pool

1. **Companies → Add company**: code, name, aliases, plan, maximum users. **Save**.
2. Open it, **Assignments → Assign shared pool…**: choose the pool and environment. **Assign pool**.
3. **Users → Create user** for each person, or create them in your Wings authentication service.
4. **Routing simulator**: the company and one username should show *Routed to* a member of the pool.

Give a company its own server

1. Make sure the server exists, is **Dedicated** or **Hybrid**, is in the right environment and passes **Test connectivity**.
2. Company → **Assignments → Assign / change server…**: primary server, optional secondary and failover policy. **Preview**, then **Apply now**.
3. If it previously used a shared pool, select **End** on that pool assignment. The dedicated server takes precedence either way, but this keeps the company's records clear.
4. Simulate one of its users.

Give one user a server of their own

1. If the server is not yet allowed for the company's data, company → **Data authorisations → Grant server**.
2. **User assignments** → find the user → **Assign dedicated server**: choose it, add a reason, **Preview**, **Assign now**.
3. Simulate the user: the source should read *User assignment*.

Take a server down for maintenance

1. Plan ahead: **Maintenance → Schedule maintenance**, or, right now: server → **Operate → Drain…** with a timeout.
2. Watch **Active sessions** on the server page fall. The server switches to Maintenance when they reach zero or the timeout passes.
3. Do the work.
4. **Operate → Restore to service**, then check it shows **Online** and **Healthy**.

Move a company to new hardware

1. Register and test the new server.
2. Company → **Transfer…** → choose the target → **1. Validate**. Fix anything that fails.
3. Choose **Scheduled** with a quiet **Cutover at** time, or **Immediate**. **2. Perform transfer**.
4. After the cutover, simulate a user and check **Routing history** for the reroutes.

Find out why a user reached the wrong server

1. **Routing simulator** with their company and username: the **Step-by-step trace** names the rule used and every server skipped.
2. **Routing history** filtered by the user: when their route last changed and why.
3. **Audit log** filtered by the company: who changed an assignment, and the reason.

Remove someone's access at once

1. **User assignments** → the user → **Disable membership**.
2. **Active sessions** → filter by username → **Revoke** each session.

Add many user assignments

1. **Bulk upload → Download template**; fill it in (section 7).
2. Upload, **Validate**, fix every row marked invalid, validate again.
3. **Commit**, then check a few users in the simulator.

14. Troubleshooting and FAQ

Start with the Routing simulator: it explains almost every routing surprise in one screen.

Troubleshooting

Symptom	Likely cause	Fix
A user always gets "company, username or password is incorrect"	Wrong company code, account locked after 5 failures, or membership disabled	Check the user in **User assignments**; use **Reset password** to clear a lockout
A whole company sees "temporarily unavailable"	Its server is down and it has no failover policy, or it has no route at all	Simulator: the reason says which. Restore the server, attach a failover policy, or give it a route
Users of one company land on different servers	The company uses a shared pool without *PerCompany* affinity, or some users have their own assignments	Simulate two users and compare the **Source** line
A change you saved does not show up	Gateways reload every 15 seconds	Wait 15 seconds, check **Configuration** shows every gateway **In sync**
A server stays Offline although it is running	It fails its health check: wrong health URL, missing health key, or the application reports unhealthy	**Test connectivity** on the server shows the failure reason
A recovered server takes no users	It is still **Recovering**, or *Require manual restore* is on	Wait about 30 seconds, or select **Restore to service**
A server never finishes draining	Users are still active on it	Wait for the drain timeout, or select **Maintenance** to move them now
You cannot pick a server in the assignment dialog	It is not authorised for the company's data	Company → **Data authorisations → Grant server**
"Changed by someone else" when saving	Another administrator saved first	Reload and make your change again

Symptom	Likely cause	Fix
You cannot retire a server	Something still uses it	The message lists the company, user or pool; move them first
A gateway shows not in sync, or a database warning	That gateway cannot reach the database	Fix connectivity; it keeps routing on its last good configuration for up to 30 minutes
Everything shows "Unknown" health	The health monitor is not running	Start the monitor service (or the admin service that hosts it)

FAQ

Can two companies have users with the same username? Yes. They are separate people with separate assignments; the company typed at sign-in decides which one signs in.

Can a user pick a server by changing a cookie or header? No. The gateway decides from its own records and the encrypted session only, and removes any identity headers a browser sends.

Do users notice a failover? Usually not: the next page simply comes from the backup server. If the application keeps state on the server, they may need to sign in again.

Do I need to restart anything after a change? No. Saved changes reach every gateway within about 15 seconds.

Why didn't my user fail over to the shared pool? Dedicated tenants only move to servers their failover policy names, so their data never ends up somewhere unapproved.

How long is history kept? See **Settings**: audit 365 days, routing decisions 30 days, metrics 90 days by default.

Who can see the backend keys? Only SuperAdministrators, and every view is recorded in the audit log as *SecretsExported*.

15. Appendix

Trying it locally

A local trial runs eight sample application servers, the admin console and one gateway on your computer, with sample companies already set up. It needs the .NET 10 SDK.

1. In the project folder, start everything: `./scripts/run-dev.sh`
2. The passwords are generated on first start and saved in `App_Data/dev-credentials.env`: `Seed__AdminPassword` for the admin console, `Seed__SampleUserPassword` for every sample user.
3. Admin console: http://localhost:5290, username `admin`.
4. Gateway: http://localhost:5280. Sign in with one of the sample users below, then open http://localhost:5280/api/whoami to see which server answered.
5. Stop everything with `./scripts/stop-dev.sh`.

Company	Username	Lands on	Shows
ABC (ABC Motors)	user1	APP-SERVER-01	A user-dedicated server
ABC	user2, manager	APP-SERVER-02	The company's server; APP-SERVER-06 is its backup
XYZ (XYZ Motors)	manager, user1	One member of Shared Pool A	A shared pool where the whole company stays on one server
GFL (Global Freight Logistics)	manager	APP-SERVER-07	A dedicated tenant with no failover
ABC	former.employee	—	A disabled account: sign-in is refused

Both ABC and XYZ have a user called `manager`: two different people on different servers.

Glossary

Term	Meaning
Affinity	Keeping a session, user or company on the same server
Alias	Another name users may type for their company at sign-in
Assignment	A dated rule that sends a user or company to a server or pool
Cluster	A company's own group of servers, used as backups or shared between its users

Term	Meaning
Data authorisation	Permission for a server to hold a company's data
Dedicated / Shared / Hybrid server	Serves one company / companies via pools / both
Drain	Stop new users on a server and let current users finish
Environment	Production, Staging, Test or Development; users stay within the environment of the address they used
Failback	Returning users to their original server after it recovers
Failover	Moving users to an approved backup server
Gateway	The service users sign in to; it forwards every request to the right server
Health check	The regular test of each server's readiness, CPU, memory and errors
Load score	A single figure combining CPU, memory, requests, response time and errors
Pool	A group of servers that companies share
Session	One signed-in browser, valid for 30 minutes idle and 12 hours at most
Simulator	The admin page that shows what routing would do, without doing it